01 / EXECUTIVE SUMMARY

Embedded Tor, ready for board bring-up

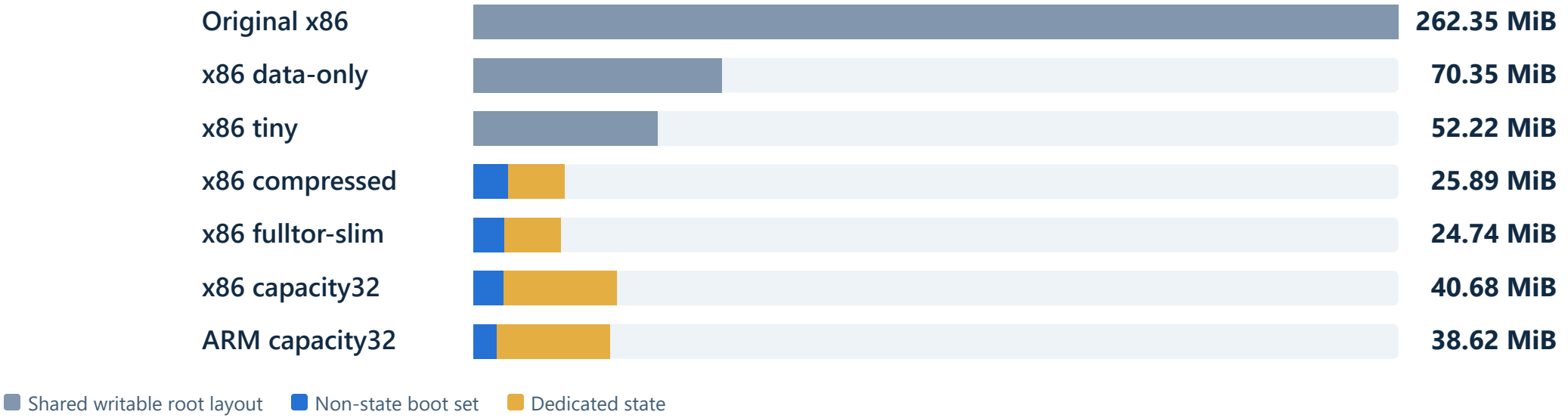
Two independent Linux endpoints for private bidirectional TCP between Windows hosts.

38.62 MiB
ARM boot set, including 32 MiB writable state

6.62 MiB
ARM kernel, read-only root and layout

256 MiB
Lowest tested ARM QEMU budget to clear every gate

02 / Optimization journey · 85.3% smaller full boot set



Logical boot artifacts only. The original development image includes its test tooling. Capacity32 uses an external fixture and deliberately reserves more writable state.

03 / STORAGE OPTIMIZATION

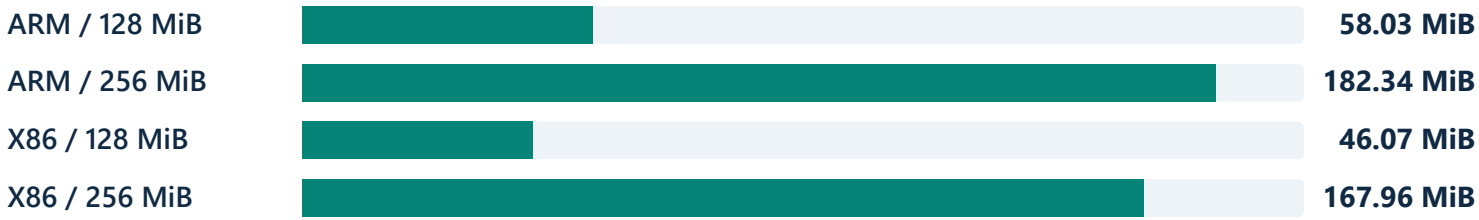
Keep state headroom while shrinking the system

Boot set (MiB)	Kernel	Read-only root	MBR + alignment	State	Total
x86 capacity32	3.76	3.92	1.00	32.00	40.68
ARM capacity32	2.37	3.25	1.00	32.00	38.62

The ARM non-state footprint is 23.71% below x86 capacity32. The 32 MiB state allocation addresses transient write failures observed on 16 MiB volumes. NAND boot, ECC, reserved blocks and update space are separate board budgets.

04 / RAM OPTIMIZATION

Minimum available RAM with the 32 MiB resident reservation



Guest	Idle PSS MiB	Tor peak RSS MiB	System peak MiB	Tor CPU s
ARM / 128 MiB	22.36	25.96	59.68	31.32
ARM / 256 MiB	22.37	26.06	62.43	31.05
X86 / 128 MiB	25.76	28.28	65.33	21.00
X86 / 256 MiB	25.63	28.23	68.69	21.60

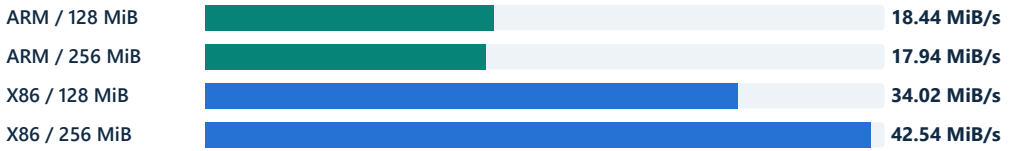
Three cold boots per row. Bars show the lowest sampled RAM margin. Table values are medians. Workload uses one target Tor and a host peer.

All groups cleared the 20% and 16 MiB absolute RAM margins

05 / PERFORMANCE AND RELIABILITY

Both 128 MiB groups missed performance gates

Local TCP



Public Tor payload



Onion connection



Public bootstrap



Guest	Bootstrap s	Onion s	Public KiB/s	Local MiB/s	Functional runs	Screen
ARM / 128 MiB	15.23	4.32	143.80	18.44	3/3	REJECTED
ARM / 256 MiB	14.84	3.15	191.67	17.94	3/3	PASS
X86 / 128 MiB	11.96	3.92	155.22	34.02	3/3	REJECTED
X86 / 256 MiB	9.29	4.01	157.70	42.54	3/3	PASS

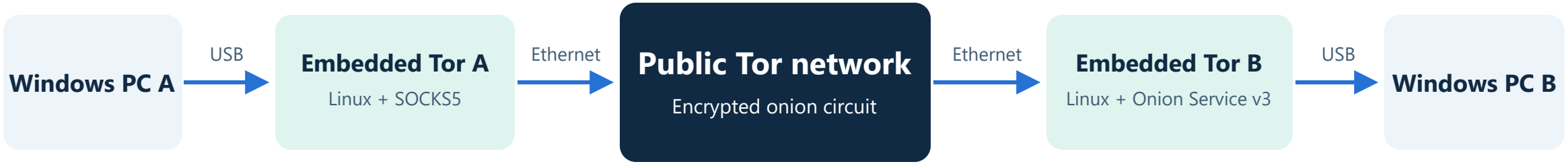
ARM 128 MiB passed all eight local gates. Public Onion latency was 4.32 s against a 3.78 s limit, and transfer was 143.80 KiB/s against a 153.33 KiB/s floor. x86 128 MiB missed local TCP and bootstrap gates.

The unchanged 20% limits compare 128 MiB with 256 MiB on the same architecture. ARM 128 MiB public throughput ranged from 90.80 to 299.40 KiB/s. Cross-ISA performance is descriptive. Earlier x86 rejections and blocked attempts remain in the technical evidence.

06 / EMBEDDED ARCHITECTURE

Tor and persistent identity live on the endpoints

USB provides local access. Each embedded Linux endpoint uses its wired uplink to reach public Tor.



Bidirectional application TCP over the onion circuit

SOCKS5

PC A directs its application TCP to the client proxy on endpoint A. The selected USB gadget must keep access local.

Onion Service v3

Endpoint B publishes an onion identity and maps the virtual port to its service. The identity persists across restart.

Physical integration

The virtual lab exercises Tor, integrity and recovery. The next phase proves the USB, Ethernet and storage paths on actual boards.

07 / HARDWARE SELECTION

Bring up DHCOM first. Retain 128 MiB as a research target.

Platform / role	CPU	RAM / flash	USB / Ethernet	Readiness and trade-off
DHCOM i.MX6ULL HI00096 First prototype	1× Cortex-A7 528 MHz	256 MiB / 128 MiB NAND + 2 MiB boot SPI	HS USB OTG and 100 Mb/s on picoITX carrier	Fits capacity target. Existing Linux SD-carrier DTB. NAND boot and gadget require integration.
100ASK T113s3-PRO V1.3 LTS Minimum-RAM candidate	2× Cortex-A7 up to 1.2 GHz	128 MiB / documented 128 MiB SPI NAND option	USB-C OTG device mode and 100 Mb/s RJ45	Smallest RAM budget. Confirm NAND variant and pin multiplexing. Needs board qualification.
MYIR MYD-YF135 Dual-Ethernet fallback	1× Cortex-A7 up to 1 GHz	256 MiB / 256 MiB NAND	HS USB OTG and 2× Gigabit Ethernet	More wired flexibility. NAND exceeds preferred budget. Board port and testing remain open.

DHCOM SKU · picoITX · 100ASK board · 100ASK NAND · MYIR SKU

08 / RISKS AND LIMITATIONS

Board performance

TCG and ISA-specific crypto affect CPU and TCP rates. Verify PHY, clocks, entropy, thermal behavior and real throughput.

USB and Ethernet

Prove simultaneous gadget enumeration, host reconnect and Ethernet uplink. Use a board-specific kernel and device tree.

Storage and reliability

Allow for cache growth, ECC and NAND wear. Exercise power loss and sustained traffic. Public Tor timings vary between circuits.

All three platforms require physical validation

09 / ROADMAP

Qualify one board before pursuing the minimum

Step	Milestone	Exit evidence
Done	ARM QEMU validation	ARM build and frozen tests at 128 and 256 MiB. Full screen passes at 256 MiB.
01	Physical board bring-up	DHCOM serial boot from removable media, DHCP, valid clock and public Tor bootstrap.
02	Boot from NAND	Board-specific flash map and identity recovery after interruption.
03	USB and Ethernet integration	Windows enumeration and reconnect while Ethernet stays operational.
04	Two physical endpoints	All nine required assertions and verified traffic in both directions.
05	Reliability and endurance	Cold boots, sustained traffic, state growth, thermal and flash endurance records.
06	Optimization and final selection	Compare 128 MiB hardware under equivalent conditions and preserve the gates.

10 / Final recommendation

Use the 256 MiB RAM / 128 MiB NAND DHCOM platform for the first bring-up. Advance the 128 MiB candidate after repeated hardware tests show acceptable latency, throughput and memory margin.

Technical handover · Roadmap · ARM measurements · x86 capacity study · Original reference · Source-linked dataset